Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Electrochemical Recycling of Lithium-Ion Batteries via Advanced Ion-Exchange and Bipolar Membrane Electrodialysis			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical and Electronics		Mechanical – Materials	
	Medium Category	Small Category	Medium Category	Small Category
	Batteries	Secondary Batteries	Clean Production	Resource Recycling Technologies
Concept of Proposed Technology	This project aims to develop a scalable electrochemical recycling platform for lithium-ion battery leachates based on advanced ion-exchange membranes and bipolar membrane electrodialysis (BMED). By selectively separating and recovering lithium and transition metals (Ni, Co, and Mn) while simultaneously regenerating acids and bases, the proposed technology significantly reduces chemical consumption, secondary waste generation, and process complexity compared to conventional hydrometallurgical recycling. The U.S. research team will provide next-generation ion-exchange membranes, electrochemical characterization, and lab-scale electrodialysis system designs, which will guide the Korean industrial partner in scaling up membrane production and developing a prototype electrodialysis cell validated using both simulated and real battery recycling streams, enabling a rapid transition toward pilot-scale deployment of a high-efficiency electrochemical recycling process for secondary batteries.			

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Rationale for Joint R&D Project	Electrochemical recycling of lithium-ion batteries using ion-exchange membranes and bipolar membrane electrodialysis (BMED) represents a strategic pathway to reduce dependence on imported critical materials while minimizing chemical use and secondary waste. However, industrial deployment is limited by gaps in membrane performance and scalable system design. This joint R&D project is essential because it combines advanced membrane and electrochemical expertise from the U.S. research team with the Korean industrial partner's capabilities in manufacturing, system integration, and validation using real recycling streams. The collaboration accelerates commercialization of high-efficiency, low-waste recycling technologies that strengthen the sustainability and global competitiveness of Korea's secondary battery industry.
Key Words	1. Electrochemical Battery Recycling 2. Bipolar Membrane Electrodialysis 3. Ion-Exchange Membrane Separation 4. Low-Waste and Circular Recycling Process 5. Scalable, High-Efficiency Cathode Metals Recovery

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	**A. Overarching Goal** The overarching goal of this R&D project is to develop next-generation ion-exchange membranes (IEMs)—cation exchange membranes (CEMs), anion exchange membranes (AEMs), and bipolar membranes (BPMs)—specifically engineered for electrochemical battery material recycling environments (**Figure 1A**). The project focuses on advancing membrane chemistry, structure, and interfacial design to achieve high ionic conductance, selective ion transport, low electrical resistance, and long-term chemical and electrochemical stability in the aggressive, multi-component leachates characteristic of lithium-ion (Li^+) battery recycling. Current commercial IEMs and BPMs (e.g., ASTOM, Fumatech) were primarily developed for applications such as seawater desalination and general chemical processing and are not designed to withstand the high acidity, high transition-metal concentrations, sulfate-rich media, chelating agents, and sustained electric fields encountered in battery recycling. As a result, membrane limitations (rather than electrochemical principles) remain the dominant barrier to efficient, scalable electrochemical recycling. This project addresses that gap by positioning membrane innovation as the foundational enabling technology, independent of any single process configuration. The developed CEMs and AEMs will be tailored to control the transport and selectivity of Li^+ and transition metal ions (Ni^{2+}, Co^{2+}, Mn^{2+}), while BPMs will be engineered to enable efficient water dissociation and reagent regeneration under recycling-relevant conditions. Together, these membrane technologies will support a range of electrochemical recycling architectures. As one illustrative example (**Figure 1B**), they can enable “zero-waste” electrochemical recovery of critical battery materials [1, 2], in which lithium is selectively recovered and acids/bases are regenerated in situ, while transition metals are recovered in forms suitable for downstream reuse. However, the primary outcome of this project is not a fixed separation flowsheet but a robust, application-specific membrane platform that can be deployed across multiple electrochemical battery recycling processes. This membrane-centric approach directly supports TIPA’s objective of core technology development and commercialization by Korean SMEs, providing transferable, scalable technologies that strengthen domestic capabilities and global competitiveness in next-generation battery recycling. **B. Addressing the "Membrane Gap** Current commercial membranes (e.g., Astom, Fumatech) are not optimized for the harsh, multicomponent nature of black mass leachate. They suffer from critical limitations when exposed to high concentrations of transition metals, acidic pH levels, and organic chelating agents (e.g., EDTA/Citrate). In standard systems, these conditions lead to rapid organic fouling, poor ion selectivity (e.g., inability to distinguish Li^+ from Co^{2+}), and membrane degradation (e.g., swelling and delamination). This project places material innovation at the center and we aim to move beyond "off-the-shelf" components to design membranes with tailored ion transport channels for high ion conductivity and selectivity, electrochemical performance, and durability required to resist aggressive chemical environment and maintain high transport efficiency in "dirty" industrial streams.				

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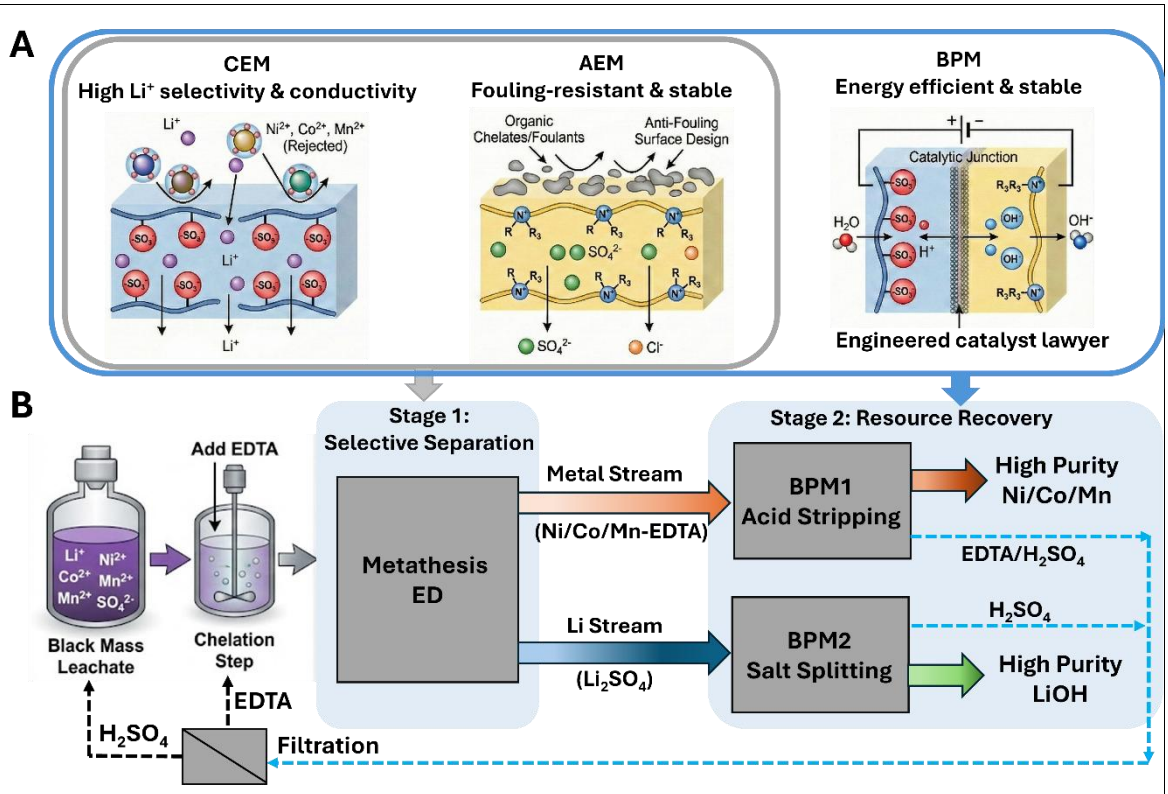

Figure 1. **A. Schematic illustration of membranes**. (Left)-Precise control and ion exchange functional groups for monovalent ion selectivity in metal-rich leachates; (middle)-high anion conductivity and engineered interface for stability against aggressive chemical & chelating agents; (right)-efficient acid/base regeneration and excellent stability.

B. Overview of Example Process: The integrated system utilizes a simplified electrochemical circuit designed to separate monovalent Li^+ from divalent transition metals (Ni^{2+}, Co^{2+}, Mn^{2+}) without the need for complex multi-stage solvent extraction. **Stage 1**: Selective Li extraction (monovalent ED). The black mass leachate is fed into a specialized ED stack equipped with monovalent selective CEM. These membranes act as a "sieve," permitting the rapid transport of Li^+ while retaining >99% of the divalent Ni/Co/Mn ions in the feed stream. **Stage 2**: Resource Recovery. *(a) BPM1: Mixed precursor recovery*. The retentate stream, now depleted of Li, contains a high-purity mixture of Ni, Co, and Mn. This stream is harvested directly as a liquid precursor (pCAM) feedstock, ready for composition adjustment and co-precipitation into new cathode active materials (Hydro-to-Cathode). *(b) BPM2: LiOH production*. The extracted Li_2SO_4 stream undergoes "Salt Splitting" in the second BPM-ED. This stage simultaneously converts the Li salt into battery-grade LiOH and regenerates H_2SO_4, which is recirculated back to the leaching tank to close the chemical loop. The EDTA will be recovered after filtration via the decomplexation of EDTA from metals by adjusting pH with H_2SO_4.

C. Specific Material Innovations

1. High performance CEMs and AEMs

We will develop highly-ion conductive/selective CEMs featuring functional groups specifically designed to facilitate Li^+ transports. This is critical for enabling the high-purity separation of Li^+ from other ion species. Simultaneously, we will engineer AEMs with anti-fouling surface functionalities capable of resisting organic fouling caused by metal-chelate complexes (e.g., $[Co(EDTA)]^{2-}$), ensuring stable operation during the chelation-assisted separation processes essential for NCM recovery.

2. Next-Generation Bipolar Membranes (BPM)

A key objective is the advancement of BPMs to enable "Salt-Splitting" technology. Current BPMs often suffer from high voltage drops and delamination at high current densities. As shown in **Figure. 2**, our proprietary BPMs have already demonstrated superior electrochemical stability and water-

splitting efficiency compared to leading commercial products (e.g., Fumatech). Building on this foundation, we will further optimize the water-dissociation catalyst layer at the membrane interface to minimize operating voltage. This optimization is vital for the energy-efficient, *in-situ* regeneration of sulfuric acid (H_2SO_4) and lithium hydroxide (LiOH) from waste sulfate streams, effectively eliminating the generation of sodium sulfate waste and closing the chemical loop.

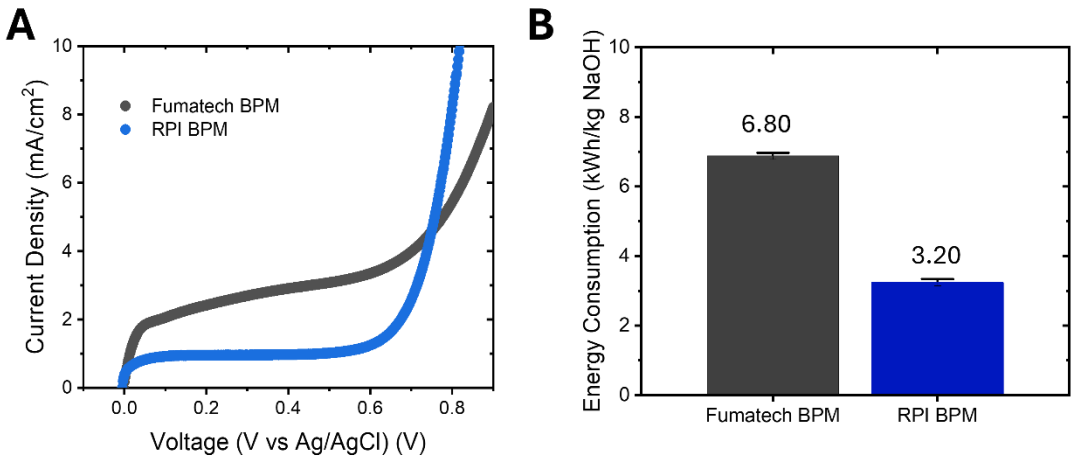

Figure 2. **A. Current-Voltage Curve of BPMs** in 1M of LiCl solution**.** Compared to the Fumatech BPM, our BPM demonstrated a higher current density at water splitting voltage and lower limiting current density. A lower water splitting voltage (or higher current density) suggests enhanced water dissociation kinetics, while a reduced limiting current density can be a favorable indicator of improved ion selectivity (i.e., minimized Li^+ and Cl^- crossover) and increased water splitting efficiency (i.e., higher Faradaic efficiency). **B. Energy consumption rate of a single BPM stack (AEM-BPM-CEM)**. A 1.5 M Na_2SO_4 solution was used as the electrode rinsing solution, while 1.5 M NaCl served as the feed solution. At 10 mA/cm², the BPM-ED with RPI-BPM achieved an energy consumption per NaOH of 3.2 kWh/kg compared to that of Fumatech BPM is 6.8 kWh/kg of NaOH.

3. Integration and Scalability (Lab-to-Fab Strategy)

Beyond material synthesis, the U.S. team will lead the integration of these advanced membranes into a unified BPM-ED system at the laboratory scale. We will validate the technology by demonstrating superior current efficiency and lower energy consumption compared to state-of-the-art commercial stacks (e.g., Astom). Finally, to ensure industrial viability, we will demonstrate the translation of lab-scale polymer synthesis into large-scale polymer production (with Orion), enabling Roll-to-Roll (R2R) continuous casting processes, validating that these high-performance membranes can be manufactured cost-effectively at dimensions compatible with commercial electrodialysis stacks.

R&D Target Specifications

The target specifications are designed to remove the principal barriers to industrial electrochemical battery recycling, specifically membrane selectivity, resistance (ASR), BPM water-splitting efficiency, durability under harsh leachates, and system-level energy/current efficiency, while ensuring scalability and cost competitiveness.

1. Ion-Exchange Membrane (IEM) Performance Target

A. CEM

- Li^+ conductivity: ≥ 5–10 mS·cm^{-1} (battery-leachate relevant conditions)
- Selectivity: Li^+ / (Ni^{2+}, Co^{2+}, Mn^{2+}) ≥ 6
- ASR: ≤ 1.5 Ω·cm² at 100–300 A·m^{-2}

B. AEM

- ASR: ≤ 0.6 Ω·cm² in acidic sulfate-rich media
- Anion selectivity: SO_4^{2-} transport number ≥ 0.95
- Stability: ≥ 3,000 h cumulative operation at pH 1–3

	C. BPM • Water dissociation voltage: ≤ 1.0 V at 100–300 A·m⁻² • Energy for acid/base generation: ≤ 1.8 kWh·kg⁻¹ acid/base equivalent • ASR: ≤ 2.5 Ω·cm² • Acid/base current efficiency: ≥ 85–90% **D. IEM Durability** • Lifetime: ≥ 1,000 h cumulative operation in acidic, metal-rich, electrochemically driven conditions • Degradation: ≤ 10% loss in conductivity/selectivity after durability testing • Fouling tolerance: stable operation using real recycling leachates (including sulfate and multivalent metals) **2. Scalability and Manufacturing Targets** • Membrane sheet size: ≥ 20 × 20 cm² (prototype-ready) • Manufacturing compatibility test by roll-to-roll or scalable batch fabrication • Cost target: ≥20% reduction versus imported commercial ED/BPM membranes (e.g., ASTOM, Fumatech) • Stack compatibility: stable operation at 100–300 A·m⁻² **3. ED/BPM-ED System (Laboratory Scale) Targets** **A. Element recovery performance** • Li recovery: ≥ 90–95% • Ni/Co/Mn recovery: ≥ 90% for each element • Product purity: ≥ 99% (battery-grade equivalent, metal basis) **B. Waste reduction and circularity** • Secondary waste reduction (e.g., Na_2SO_4): ≥ 50% vs. conventional routes • Acid/base reuse via BPM-ED: ≥ 90% reagent recycling rate **C. Process efficiency and stability** • Energy consumption: ≤ 2 kWh·kg⁻¹ recovered lithium equivalent • Stable continuous operation: ≥ 300 h using simulated or real streams **4. Comparison to State-of-the-Art (SoA) Commercial ED Systems** Recent multi-stage ED demonstrations for LIB recycling (using standard commercial membrane types in a PCCell electrodialyzer) report baseline membrane resistances on the order of ~2–6 Ω·cm² depending on membrane type (e.g., ~2.5 Ω·cm² for a sulfonic-acid CEM, ~2 Ω·cm² for an AEM, and ~6 Ω·cm² for a monovalent-selective CEM) [1]. These SoA configurations also highlight operational challenges such as membrane fouling, dilute product streams, and low current utilization that reduce overall efficiency and increase operating cost. Besides, SoA emphasizes that process efficiency is constrained by BPM energy demand, current efficiency, and fouling/degradation in real solutions, making BPM performance a gating factor for closed-loop recycling.
R&D Plan	The R&D plan is structured to systematically develop, validate, and scale ion-exchange membranes (CEM, AEM, and BPM) as the core enabling technologies for electrochemical recovery of lithium and transition metals (Ni, Co, Mn) from spent lithium-ion battery leachates. The approach integrates membrane materials design, electrochemical characterization, lab-scale ED/BPM-ED system evaluation, and scalability assessment, ensuring that membrane innovations directly translate into system-level performance improvements. **Task 1. Battery Leachate Definition and Baseline ED/BPM-ED Performance (RPI/UIC)** • Define representative battery leachate compositions (Li, Ni, Co, Mn, sulfate, acidity), including simulated and real recycling streams • Establish baseline ED and BPM-ED performance using commercial membranes

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- Identify dominant limitations related to membrane properties such as membrane resistance and selectivity, current efficiency losses, fouling and stability in metal-rich solutions.
- Generate benchmark data for comparison against newly developed membranes

Task 2. Development of Advanced CEM and AEM for Multi-Metal Separation (RPI)

- Design and synthesize CEMs optimized for high Li^+ conductivity and controlled selectivity between monovalent and divalent cations.
- Design and synthesize AEMs optimized for high sulfate conductivity, suppressed transport of metal-chelate complexes, and improved current efficiency in ED and BPM-ED process.
- Characterize membrane properties: Ion conductivity and transport numbers, ASR, and chemical and electrochemical stability in acidic leachates

Task 3. Bipolar Membrane (BPM) Design and Optimization (RPI)

- Develop thin (<70 μm) BPMs with chemically compatible CEM/AEM interfaces, reduced ohmic resistance, and enhanced interfacial adhesion to prevent delamination
- Optimize AEL and CEL polymer chemistry and interfacial morphology
- Optimize water-dissociation efficiency and BPM stability under acidic, metal-rich environments
- Evaluate BPM performance for water dissociation voltage, acid/base generation efficiency, and salt leakage and durability

Task 4. Lab-Scale ED and BPM-ED Stack Assembly and Evaluation (UIC)

- Assemble lab-scale ED and BPM-ED stacks incorporating developed CEMs, AEMs, and BPMs
- Evaluate system performance under controlled conditions such as metal recovery efficiency (Li, Ni, Co, Mn), current efficiency and energy consumption, and acid/base regeneration rate
- Optimize operating parameters such as current density, flow configuration, and feed concentration and pH

Task 5. Modeling, Optimization, and Durability Assessment (UIC)

- Develop 1D and 2D numerical models (COMSOL-based) for ion transport and polarization, loss mechanisms (leakage, co-ion transport), and guidance of stack and membrane optimization
- Conduct long-term durability testing using real battery recycling streams
- Refine system design based on feedback from modeling and experiments

Task 6. Scalability, Manufacturing Readiness, and Pilot Relevance (RPI/UIC)

- Demonstrate scalable membrane fabrication (sheet size ≥30 × 30 cm^2)
- Validate compatibility with pilot-scale ED/BPM-ED stacks
- Perform preliminary techno-economic and process assessments to evaluate cost reduction potential, waste minimization, and energy efficiency gains

Overall Timeline Summary

- **Year 1:** Membrane design, baseline benchmarking, initial BPM development
- **Year 2:** Integrated ED/BPM-ED validation, modeling-guided optimization
- **Year 3:** Durability testing, scale-up validation, pilot readiness

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	**A. Technical Leadership and Core Innovation** Rensselaer Polytechnic Institute (RPI) will serve as the technical lead and primary innovation driver for this R&D project, with a central focus on advancing ion-exchange membrane technologies for electrochemical lithium-ion battery recycling. The RPI will lead both fundamental and applied research on next-generation CEMs, AEMs, and BPMs. **B. Research Scope and Key Technical Tasks** Within the defined research scope, the RPI will design, synthesize, and modify ion-exchange membrane materials with controlled polymer chemistry, ion exchange capacity, functional groups, and morphology to enable selective recovery of lithium and transition metals (Ni, Co, Mn). The institute will conduct comprehensive electrochemical and physicochemical characterization, including ionic conductivity, area-specific resistance, transport numbers, selectivity, water-dissociation performance (for BPMs), and chemical and electrochemical stability under recycling-relevant conditions. These activities will provide the scientific basis for membrane down-selection and iterative optimization. **C. Process Development and System Evaluation** University of Illinois-Chicago (UIC) will design, assemble, and operate lab-scale ED and BPM-ED systems incorporating the developed membranes. This includes systematic investigation of operating parameters such as current density, voltage, feed composition, as well as quantification of metal recovery efficiency, energy consumption, current efficiency, and acid/base regeneration performance. The UIC will also support modeling and data analysis efforts to link membrane properties with system-level performance and to guide scale-up considerations. **D. Knowledge Transfer and Collaboration** A critical responsibility of RPI will be to ensure effective knowledge transfer and close technical collaboration with the Korean SME partner. This includes sharing membrane formulations, fabrication protocols, performance data, and design guidelines to support membrane scale-up and prototype system development. The U.S. team will participate in regular technical meetings, joint data reviews, and collaborative problem-solving activities and will contribute to technical reporting, milestone assessments, and dissemination of non-confidential results as appropriate.
Expected R&R of Korean SME Partner	**A. Industrial Scale-Up Leadership (Lab-to-Fab)** The Korean SME will serve as the Project Coordinator and Industrialization Lead, bridging the gap between RPI's lab-scale innovations and mass production. Specific task is to establish a Roll-to-Roll (R2R) Process capable of producing CEM, AEM, and BPM sheets with a width of 50 cm. This involves optimizing solvent evaporation rates, coating speeds, and substrate tension to ensure uniform thickness and defect-free production. **B. System Integration & Prototype Fabrication** The SME is responsible for the engineering design and assembly of the Pilot-Scale BPM-ED Stack; they will incorporate the custom-manufactured membranes into the stack, ensuring leak-proof sealing and mechanical stability under high osmotic pressure differentials typical of high-concentration battery leachates. **C. Performance Validation (Real-World Testing)** • Test Bed: Operate the pilot BPM-ED system using real black mass leachate sourced from domestic battery recyclers. • Validation: Conduct 1,000-hour continuous operation tests to evaluate membrane durability, fouling resistance, and the purity of the recovered LiOH and regenerated H2SO4. **D. Commercialization & Supply Chain Strategy** • Cost Analysis: Perform a detailed TEA to demonstrate that the new membrane production cost is at least 30% lower than imported equivalents (e.g., Astom, Fumatech). • Market Entry: Develop a business model to supply these specialized membranes to the growing Korean battery recycling market, securing domestic supply chain independence for critical recycling components.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**A. Technological Competitiveness (Securing "Super-Gap" Technology)** • Establishment of Proprietary Membrane IP: Through this Global Collaborative R&D, we will secure original intellectual property for "Battery-Recycle Grade" IEMs that surpass the performance of current Japanese and German incumbents (e.g., Astom, Fumatech). • Realization of a Zero-Waste Platform: We will successfully demonstrate a closed-loop BPM-ED system capable of recovering >90 % of Li while regenerating acid/base reagents. This overcomes the "Sodium Sulfate Waste" bottleneck that currently plagues the hydrometallurgical recycling industry. • Technological Internalization: By transferring RPI's advanced polymer synthesis protocols to the Korean SME, we will elevate domestic membrane manufacturing technology to narrow the technology gap with advanced IEM development technology. **B. Economic and Industrial Impact** • Import Substitution & Supply Chain Security: Currently, Korea relies heavily on imported IEMs. This project will enable the localization of these core materials, reducing dependency on foreign supply chains and lowering procurement costs. • Cost Leadership in Battery Recycling: By eliminating the continuous purchase of H2SO4 and NaOH and avoiding waste disposal fees, the proposed technology is projected to reduce the Operating Expenditure (OPEX) of battery recycling plants by >20 %. • New Market Creation: The Korean SME will expand its business portfolio from "Equipment Assembly" to "High-Value Material Manufacturing," opening export opportunities for both Membrane Rolls and Turnkey BPM-ED Recycling Units to global gigafactories in the U.S. and EU. **C. Social and Policy Contribution** • Response to Environmental Regulations: The technology provides a direct solution to tightening regulations (e.g., EU Battery Regulation 2023, US Inflation Reduction Act) that mandate high recycling rates and lower carbon footprints. The "Zero-Waste" process significantly reduces the industrial discharge of saline wastewater. • Resource Security: By improving the recovery rate of critical minerals (Li, Ni, Co) from domestic waste streams, this project contributes to National Resource Security, reducing Korea's vulnerability to raw material price volatility.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization